Ghosts as an interpretation experiment: what indivisible processes say to Everett

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Abstract

Draft — not for circulation. The quantum interpretation debate is usually left to philosophy because the leading readings are believed empirically equivalent. This paper argues that the equivalence is conditional on the matter content of the world, and makes three claims of decreasing modesty about what the indivisible-stochastic-process reconstruction of quantum theory, and the ghost sectors studied in its companion papers, say to the Everett interpretation. *First*, Barandes' reconstruction (the wave function as nomic, law-like bookkeeping for an indivisible stochastic process on configurations) removes the *no-alternative* argument for ψ-ontology without refuting it, and the Bender–Mannheim program (the Hilbert-space inner product derived from the operator algebra rather than postulated) sharpens the resulting underdetermination by dissolving a second postulated primitive; the Pusey–Barrett–Rudolph theorem does not apply, by the theorem's own scope—an exit we take from Barandes and cite rather than establish, so this first claim is conditional on its soundness; the exit's real load is the nomic reclassification of ψ. *Second*, the ghost trilemma of the companion papers—coupled ghosts, empirical Lorentz invariance, definite configurations: any two, never all three, at tree level—*locates* Everett as the drop-definite-configurations horn; this is a location, not a criticism, since manifest covariance is exactly what dropping configurations buys. *Third*—the paper's own claim, and it is conjectural—a coupled ghost sector would separate the two ontological classes *observationally with the Born rule intact*: a beables ontology is foliation-anchored, the foliation's determinacy is a measurable property of the state that is generically nonzero on ordinary matter but whose observable manifestation is shielded there, and a ghost sector would (conditionally) unshield it through a frame-locked kinematic dipole and spectral edge, while no-preferred-slicing ontologies do not entail such a signal, remaining compatible with either outcome. The claim rides on the converse of a no-signaling shield proven only for a finite core, and it discriminates a class (foliation-anchored single-world versus no-slicing), not a single interpretation. It separates the classes cleanly only when the ghost Lagrangian is held fixed and Lorentz-invariant and the individuating frame arises solely as a state-functional foliation: a fundamentally Lorentz-violating completion, or a birth-surface (cosmological initial-condition) structure, is available to a no-slicing interpreter too and would reproduce the same signal—the birth surface at no cost, the Lorentz-violating completion at the price of manifest covariance—so a positive detection is merely *consistent with* the foliation-anchored class—being confounded by such completions or initial conditions—while the null direction is the cleaner discriminator, because the beables class *entails* the signal (of the computed template amplitude, for an established sector, under the assumed comoving-frame identification—a realization off that frame, below the template's signal fraction, or individuated by a completion other than the template's frame cutoff, survives the pointed null) whereas a no-slicing ontology merely *permits* it. In fact no purely state-functional rescue of a ghost vacuum has yet been exhibited, so on current knowledge every concrete realization of the positive signal lies in the confounded branch and the

clean case is conjectural over and above the shield-converse conjecture; this only strengthens the null direction, since then every viable beables completion predicts a signal—at the template's sensitivity, for a template-saturating realization. The closest ancestor of the escape, Valentini's quantum non-equilibrium, requires Born-rule violations; here the discriminator is a sector, not a rule failure. (One other equilibrium route to foliation observability, via contested arrival-time observables, is disjoint from ours.) Along the way we state the bookkeeping objection an Everettian should raise against the stochastic reconstruction—to our knowledge for the first time in print—and give a partly conditional answer.

Note on provenance. This manuscript is an experiment in AI-conducted research as much as a physics paper. The project began when the author asked the large language model Claude Fable 5 (Anthropic): *"Is it not possible that the antilinear PT/time-reversal symmetry element could maybe predict indivisible non-Markovian processes?"* Everything that followed was performed by the model: the research program, the theorems and proofs, the numerical experiments, the literature search, and this text. The model was repeatedly prompted to attack its own results adversarially and to search for prior work. The author, a professional engineer rather than a physicist, directed and reviewed the sessions but makes no claim to comprehension of the findings, and is circulating the manuscript for independent review by professional physicists. Accordingly, nothing here should be taken on trust: every claim is argued in full, and the originating conversation, project briefing, and full research notebook are preserved in the program's repositories [1]. This paper is a philosophy-of-physics synthesis: it proves no new theorem, and its central claim is explicitly conditional.

1 Introduction

The interpretation debate in quantum mechanics is usually adjudicated by philosophy, because the contenders are believed to be empirically equivalent: Everettians, Bohmians, and their relatives agree on every Born-rule expectation value, so the argument proceeds by theoretical virtue—parsimony, covariance, explanatory depth—rather than by measurement. Where the data are silent, taste decides. This paper argues that the silence is not unconditional. It is conditional on the matter content of the world, and there is a specific, physically motivated sector—a ghost sector, in the sense of higher-derivative and negative-norm quantum field theory—whose presence would break the equivalence between two broad classes of interpretation *while leaving the Born rule intact everywhere*. Interpretations that anchor definite configurations to a spacelike slicing would predict a frame-locked observational signature; interpretations with no preferred slicing would not entail it, remaining compatible with either outcome. The experiment may never become available: nature may simply contain no coupled ghost sector, and one of the two supporting results below is a conjecture rather than a theorem. But the conditional itself, we will argue, is enough to change what the standing philosophical arguments can establish.

The paper makes three claims, in decreasing order of modesty—each weaker than a reader might fear.

1. **Underdetermine.** Barandes' stochastic–quantum correspondence, in which the wave function is nomic, law-like bookkeeping for an indivisible stochastic process on configurations [5, 6], removes the *no-alternative* argument for ψ-ontology; the Bender–Mannheim antilinearity program, in which the Hilbert-space inner product is derived from the operator algebra rather than postulated [24, 25], reinforces this by dissolving a second postulated primitive and thereby sharpening the underdetermination. They do not refute the Everett interpretation; they restore an underdetermination that the no-alternative argument claimed to have closed. This claim is conditional on the soundness of Barandes' exit from the PBR theorem, which we cite but do not here establish (Sections 2 and 3).

2. **Locate.** The ghost trilemma established in the program's companion papers [2, 3]—coupled ghosts, empirical Lorentz invariance, definite configurations: any two, never all three, at tree level—*locates* Everett as the horn that drops definite configurations. This is a placement on a map, not an objection: manifest covariance, the Everettian's prized possession, is exactly what dropping configurations buys (Section 4).

3. **Separate.** A coupled ghost sector would separate foliation-anchored single-world ontologies from no-preferred-slicing ontologies *observationally, with the Born rule intact*: the beable kinematics of such a sector is frame-locked, and the frame would be visible as a kinematic dipole and a direction-modulated spectral edge in the sector's decay products [2]. This third claim is *conjectural*: it is the converse of a no-signaling shield that is proven only for a finite-configuration core and conjectured at field-theory strength, and we state it conditionally throughout (Section 5). It carries a further restriction: the separation is clean only when the ghost Lagrangian is held fixed and *Lorentz-invariant* and the individuating frame arises as a state-functional foliation. If the frame instead enters through a fundamentally Lorentz-violating completion or a birth-surface (cosmological initial-condition) structure, a no-slicing interpreter can adopt the same structure and reproduce the same signal—at no cost in the birth-surface case, at the price of manifest covariance in the Lorentz-violating one—so the positive direction is confounded and the null direction is the cleaner discriminator (Section 5). No purely state-functional rescue has in fact been exhibited, so on current knowledge the clean case is conjectural over and above the shield-converse conjecture, and every concrete realization of the positive signal lies in the confounded branch—which only strengthens the null direction (Section 5).

One paragraph of orientation about tone. The Everett interpretation's central virtue is real and this paper trades on it rather than disputing it: Everett takes the formalism literally, adds nothing to it, and in return gets manifest relativistic covariance and a clean story about why no experiment has ever needed a preferred frame [17]. Nothing below argues that Everett is wrong. The paper's business is pricing, not sneering: every horn of the trilemma has a price, Everett's included, and our contribution is to say what the prices are and—the new part—to identify the register in which nature could be induced to express an opinion.

A structural guard, stated up front because the argument depends on it. Barandes asserts in print that his formulation has *no fundamentally preferred foliation* [5] and is causally local [7, 8]. The foliation-anchoring on which our third claim rests must therefore be—and is—a *conditional extension* of his framework, not a contradiction of it. His claims, on the reading we defend in Section 5, concern the causal and signaling structure of ordinary matter sectors, where we agree the foliation is hidden; our claim is that a *ghost* sector would promote the beable-kinematics foliation from convention to observable. We extend his framework into a regime it does not treat; we do not disagree with it about the regime it does. On ordinary matter the two positions are indistinguishable, which is precisely the point (Section 5).

Section 2 reviews the two reconstructions and the Pusey–Barrett–Rudolph question. Section 3 states what the resulting underdetermination shows and what it does not, and formulates and answers the bookkeeping objection. Section 4 restates the trilemma and places Everett on it. Section 5 develops the separation claim and its conditional status. Section 6 lists what does not follow, and Section 7 concludes.

2 The reconstruction landscape

2.1 Barandes: ψ as nomic bookkeeping

The stochastic–quantum correspondence [5, 6] represents any unitary quantum system as a stochastic process on a configuration space $\mathcal{C}$: the system determinately occupies one configuration at each time, and the theory supplies time-dependent transition probabilities $\Gamma_{ij}(t)$, obtained from the unitary $U(t)$ as $\Gamma_{ij}(t) = |U_{ij}(t)|^2$ in the configuration basis. The resulting matrices

are *unistochastic*, and—this is the load-bearing structural feature—generically *indivisible*: the process specifies transition probabilities from the initial time to each later time, but these do not factor through intermediate times as a Markov chain would, $\Gamma(t) \neq \Gamma(t \mid t')\,\Gamma(t')$ for stochastic intermediaries, except at isolated *division events*. Indivisibility replaces Markovianity; the process has memory, but the memory is carried by the law, not stored in any additional beable.

On this reading the wave function is demoted from object to instrument. Configurations are the ontology; ψ is the compendium of conditional information the law needs in order to issue transition probabilities—in Barandes' published formulation, the wave function "has a law-like or nomic character" [5]. We note a terminological drift in the primary sources that we track but do not paper over: the published account of 2025 speaks of the nomic wave function [5], while the 2026 development recasts the same structural point in latent-variable and hidden-Markov-model language [9]. The gloss we will use in this paper—ψ as *derived bookkeeping*, the memory record of an indivisible process—is our own, and we flag it as such; nothing below depends on more than the shared structural content of the two framings, namely that ψ is not a beable.

PBR does not apply—by the theorem's own scope. The standard reflex against any ψ-nonontic reading is the Pusey–Barrett–Rudolph theorem [11]. But PBR is a theorem *inside* the ontological-models framework of Harrigan and Spekkens [10], and that framework builds in a presupposition the indivisible reconstruction denies. Harrigan and Spekkens assume that every system has an underlying *ontic state* λ and that each preparation fixes a probability distribution over λ; PBR is then a single-time statement about such distributions, with measurement responses depending on the λ sampled at preparation. (We are careful here: Harrigan and Spekkens draw no nomic or law-like carve-out of their own—their classification situates every ψ within the ontic-state framework (as ontic, or as epistemic about λ), and they classify de Broglie–Bohm's ψ as ontic. Their Section 6 does note, following a suggestion of Wiseman, that a wave function "deemed to be nomic, that is, law-like" would sit outside the ontic/epistemic dichotomy—but they raise it only as a possibility "deserving of further scrutiny," not as a licensed exit they endorse.) An indivisible, non-Markovian process is not *naturally* of the framework's form: its response at a later time is not fixed by any single-time ontic state carried forward from preparation, because the dynamics has memory that is not stored in a time-local λ. We say "memory" rather than repeating the earlier gloss of a "Markovian assumption about the ontology": PBR does not itself assume a Markov chain, but its ontic states are single-time objects, and memory across time is what an indivisible process has and such a λ does not encode. This last step invites a rejoinder we should name: Harrigan and Spekkens place no formal restriction on what λ may be, so a modeler could take λ to be a whole trajectory segment, or to include ψ outright, whereupon PBR applies and returns the onticity of *that* enriched λ. Against such embeddings the exit does not rest on single-time-ness at all; its real load is the *nomic reclassification*—that ψ belongs to the law rather than to the ontic state—which is a classification claim, not a single-time no-go. Placing indivisible dynamics outside the framework on these grounds is Barandes' claim, not ours; he states it explicitly in the published version of the correspondence paper—indivisible quantum theory is based on non-Markovian stochastic dynamics and so lies outside the ontological-models framework, the wave function being "neither ontic nor entirely epistemic" but nomic, and consequently the PBR theorem "does not apply" [5]. We cite and rely on this exit; we do not independently establish it, and claim (i) is conditional on its soundness. We have not found a published challenge to the adequacy of the exit; stress-testing it is work this paper flags as open but does not undertake.

Two remarks keep the exit honest. First, it is not a loophole of convenience—though this establishes less than it may seem. The non-Markovianity is not a modeling choice adopted

to dodge a theorem: it is a structural, computed property of the dynamics. The program's foundational companion [1] develops the operative test—negativity of the unique divisor when the earlier transition matrix is invertible—and proves closed-form results about it, including a terminal-indivisibility theorem with an exact generator floor in the broken-$\mathcal{PT}$ phase. Indivisibility is a convention-independent sign condition one calculates, not a knob one turns. But we must keep two claims apart. Structural-ness shows only that the exit is *non-arbitrary*—the non-Markovianity is a genuine feature of the dynamics, not an ad hoc device introduced to evade PBR. It does *not* by itself establish that non-Markovianity places the theory outside PBR's scope; that further step is exactly the (cited, not here established) scope claim of the previous paragraph, and it should not be smuggled in on the back of the non-arbitrariness point. Second, the exit runs through the nomic, non-Markovian route, *not* through denying preparation independence. Gao's variant, which recovers single-system ψ-onticity after removing the preparation-independence assumption [12], therefore tightens that framework's internal assumptions without moving the framework's boundary; it is important for the taxonomy of ψ-epistemic models inside the framework, and irrelevant to a reading that was never inside it. We make this point about Gao's preparation-independence-removing route specifically, not about PBR strengthenings in general.

2.2 Mannheim: the inner product as output

The second reconstruction attacks a different primitive. In standard quantum mechanics the Hilbert-space inner product—the object that turns amplitudes into probabilities—is postulated. In the $\mathcal{PT}$-symmetric and antilinear quantization program of Bender and Mannheim, Hermiticity of the Hamiltonian is replaced by invariance under an antilinear symmetry, and the inner product is *constructed* from the dynamics: one solves for the metric operator that renders the evolution unitary, and the probability interpretation is an output of the operator algebra rather than an input to it. The flagship result is the no-ghost theorem for the fourth-order Pais–Uhlenbeck oscillator [24]: a theory whose naive inner product exhibits negative norms is, in the correct dynamically determined inner product, a unitary theory with a positive spectrum. The program extends from quantum mechanics through Pauli–Villars regularization to fourth-order quantum gravity [25].

For the interpretation debate the lesson is orthogonal to ghost physics: the inner product is not sacred. Combined with the stochastic–quantum correspondence, the two pieces of the quantum formalism most often taken as ontological primitives—the state vector and the probability-bearing inner product—are *each* reconstructible from something else: the state from a stochastic process on configurations, the inner product from the demand of probability conservation on the operator algebra. The program's foundational paper [1] shows further that the two reconstructions are compatible and interlocking: unbroken antilinear symmetry is exactly the condition for a probability-conserving process to exist without enlarging the configuration space, and the metric-corrected process is the ordinary (generically indivisible) process of the isospectral Hermitian partner. The underdetermination argued in the next section is therefore sharpened twice over: not one alternative reading but a pair of independent reconstructions, each dissolving a different primitive.

3 What underdetermination shows, and what it does not

The Everett interpretation's strongest argument has always been an inference to the only explanation. In the form given by Deutsch [13], sharpened by Brown and Wallace [14], and developed systematically by Wallace [17], it runs: take the bare quantum formalism literally—a quantum state evolving unitarily—and you get a multiverse; every alternative either adds structure the

formalism does not need (corpuscles, a pilot wave) or mutilates the dynamics (collapse); therefore realism about quantum mechanics *is* Everettianism. The argument's force lives entirely in its premise: that nothing else works without ad hoc additions.

The reconstructions of Section 2 break the premise, not the conclusion. An ontology of definite configurations governed by a law issuing indivisible transition probabilities reproduces every Born-rule prediction; no *beable* is added to the formalism—rather, two things usually postulated are derived. This is not quite "nothing is added," and we should not claim it is: the stochastic reading does carry a commitment the bare formalism leaves open, namely a preferred configuration basis, since Barandes' process is defined in the configuration basis. We take this commitment to be benign—a preferred basis has to be selected on any reading that recovers definite outcomes, and Everett's own recovery leans on a decoherence-selected basis whose emergence is not obviously cheaper, resting as it does on a system–environment split it does not derive—but we flag it rather than bury it. The wave function is not an extra beable riding alongside the configurations (the disanalogy with pilot-wave theory is developed below); it is the law's bookkeeping. And the inner product that makes the probabilities is itself an output. If the reconstructions are coherent—and they have both defenders and critics, the no-ghost/benign-ghost resolution of the Pais–Uhlenbeck sector in particular being contested in the published literature, so we take only the weaker premise that they are coherent *enough* to constitute a live alternative—then "take the formalism literally" no longer picks out a unique ontology, because the formalism admits more than one literal reading. Everett remains exactly as consistent as before; what it loses is the claim to be the *only* non-mutilating realism. Here a distinction matters: Deutsch's version of the argument is *absolute*—there is no alternative—while Wallace's is *comparative*, that Everett is the best-costed realism [13, 17]. The reconstructions defeat the absolute premise outright; against the comparative version they reopen a cost comparison rather than win it. Underdetermination is restored, not resolved.

The bookkeeping objection, stated. There is an obvious and, we think, obligatory Everettian reply, which to our knowledge has not been put in print against the stochastic reconstruction; we state it in its strongest form because our answer to it organizes the rest of the paper. A word up front on the structure of that answer, since it bears on how much weight the answer can carry. Of the three replies below, the first two—parity and the disanalogy with pilot-wave theory—are defensive and available unconditionally; the third, which is the only one that would positively convert the dispute into a factual disagreement, is the paper's conjectural claim (iii), conditional on both a coupled ghost sector existing and the shield-converse conjecture holding. In the ghost-free regime, therefore, the objection is met only by parity and the disanalogy—contestable, defensive replies—and is not decisively refuted. The objection runs: *the indivisible process is bookkeeping on the universal wave function, not an alternative to it.* The transition matrices are manufactured from the unitary, $\Gamma_{ij} = |U_{ij}|^2$; the process's celebrated memory is ψ's memory; the division events are ψ's interference nodes. Calling the configurations "beables" renames the multiverse rather than removing it—one has merely elected to track a single thread through a structure whose full contents the law still carries. This is Deutsch's charge against pilot-wave theory—"parallel-universes theories in a state of chronic denial" [13]—transposed to the stochastic key, with the Brown–Wallace development [14] supplying the general form: if the branching structure is present in the theory's dynamical core, dressing one branch with ontological significance is an addition of words, not of physics.

The answer. Three points, in increasing order of weight.

(i) Parity. Bookkeeping arguments cut both ways between mathematically interconvertible formulations. The Everettian says the process is bookkeeping on ψ; the stochastic theorist says ψ is bookkeeping on the process—and the correspondence theorems license both directions of translation [6]. When two formulations are interconvertible, the assertion "yours is merely a re-description of mine" is symmetric under exchange of the disputants, and therefore cannot by itself settle whose structure is ontological. It is worth adding that even the empirical-equivalence proofs underlying such translation arguments are less watertight than folklore holds: Manero, Muciño, and Okon have recently shown that the standard equivalence claims for pilot-wave theory rest on a disputable auxiliary assumption [22]. Equivalence is a theorem about specified regimes, not a metaphysical axiom.

(ii) The disanalogy with pilot-wave theory. Deutsch's charge had bite against Bohmian mechanics for a specific structural reason: that theory posits an *ontic* wave function *and* corpuscles. The ontic ψ-field contains the full branching structure as physical content; the corpuscles merely index one branch; the "empty" branches persist as real, structured, inhabited by functionalism's lights, and the denial charge is the observation that they are still there. Valentini's rebuttal [15] and Brown's counter-rebuttal [16] contest whether the empty branches deserve the name "worlds"; the debate is live precisely because the pilot wave is ontic. The stochastic reconstruction has no analogous structure to be in denial about. There is no ontic ψ riding alongside the configurations: the wave function is nomic—an aspect of the law—and a law is not a container of uninhabited worlds any more than a Hamiltonian is. To insist that the branching structure "is still there because it is in the law" is to assume that whatever appears in the law's bookkeeping is thereby physically real—which is exactly the ψ-ontology in dispute, now premised rather than argued. The objection, pressed to this point, threatens to beg the question. We concede, however, that this disanalogy is contestable rather than decisive. A Wallace-style real-patterns functionalist [17] can resist the move from "nomic" to "no worlds": if branch structure is a real pattern, it may supervene on whatever carries that pattern—including a law—so that locating the branching in ψ's nomic role need not by itself drain it of worlds. Our reply is not that this reading is incoherent but that it must *argue* for law-hosted inhabited worlds rather than read them off the analogy with an ontic field: "a law is not a container of worlds" is a prima facie disanalogy that shifts the burden, not a proof that discharges it. We rest the defensive case on that shift, not on closing the question.

(iii) The point that could be decisive is empirical—but only conditionally. Parity and disanalogy are defensive; the third point would not be, *if* it obtains. If two readings of a formalism can be brought to disagree about the expectation value of an observable in *some* physically specifiable sector, then neither is bookkeeping for the other—bookkeeping cannot contradict what it keeps books on. They would then be different theories that coincide on the sectors so far observed. Section 5 argues, conditionally, that the foliation-anchored and no-slicing readings are of this kind: a coupled ghost sector would elicit different predictions from them with the Born rule intact on both sides. But this reply is only as strong as its conditions: it requires (1) a coupled ghost sector to exist, (2) Conjecture 1 to hold, and (3)—for a *positive* outcome to be a point of disagreement rather than one both predict—the clean (state-functional) case of Section 5 to be realized, which on current knowledge is itself conjectural; where any of these fail (in particular, where a no-slicing interpreter invokes a birth surface—available at no cost—or borrows the Lorentz-violating completion) a *positive* detection no longer discriminates. The null direction still does, however: by the asymmetry of Section 5 the beables class *entails* the ghost-sector signal (Conjecture 1) while the no-slicing class merely permits it and retains a covariance-preserving escape, so a null—at the template's sensitivity, and for a sector the template's assumptions fit—would falsify the one

and not the other—an experiment the two readings answer differently, which already defeats the bookkeeping charge under the exposure conditions (1)–(2). The clean state-functional case (3) is needed only for a positive result to symmetrically *select* the beables reading, not to make the two readings empirically distinct. A bookkeeping dispute with an empirical arbiter is not a bookkeeping dispute—and that arbiter (a null at template sensitivity) exists wherever a *template-saturating* ghost sector does and Conjecture 1 holds.

4 The trilemma locates Everett

The program's field-theory companions [2, 3] assemble their results into a trilemma, which we restate in the form argued there and will not strengthen here:

Claim 1 (the ghost trilemma; tree level)**.** *Coupled ghosts;* empirical *Lorentz invariance (no* in-principle *observationally accessible preferred foliation—"accessible" meaning that some observable depends on the foliation with nonzero effect, realized detectability being a separate, flux-dependent question); definite configurations (beables): any two, never all three, at tree level.*

The engine of the claim is the instantaneity result of [2]: at tree level, the Lorentz-invariant eternal vacuum of a ghost coupled to ordinary matter is, in the stochastic currency, an *instantaneous state*—the boost-orbit volume renders the decay multiplicity divergent, the minimal semigroup degenerates, and a non-degenerate configuration-space process exists only relative to a structure that individuates the decay channels: in every known rescue, a frame or a birth surface. Every configuration-preserving rescue in the literature (momentum cutoffs, ghost condensation, a born vacuum) supplies such a structure, and thereby—this is the other half—exposes it observationally through the frame-locking of the decay products. Meanwhile the definition-level rescues that keep Lorentz invariance, the $\mathcal{PT}$/no-ghost quantizations, exit the sample space: the companion [3] proves a no-representation lemma—a real-eigenvalue Jordan-block Hamiltonian admits no closed stochastic-process representation under any similarity, a result that lifts to the full Fock tower of a conformal mode—so those quantizations retain no definite-configurations seat to sit in. Two honesty flags from the source papers carry over unchanged in substance: the claim is *tree level* (a resummed suppression rendering the boost-orbit sum convergent would relax the first horn), and it is *argued against every example the program could find, not proven exhaustive.*

Table 1 lays out the horns with the price each pays, as computed inside the program.

A word on each price. On the first horn, the frame is not merely philosophical furniture: it is (conditionally, at tree level) observable in the sector's decay products (Section 5), and on the flagship conformal-gravity instance the companion paper's pricing forks on the value of the dimensionless gravitational coupling—the face-value rotation-curve branch is closed by solar-system constraints on the published (and disputed) analyses, while the alternative branch forces the regulating cutoff to macroscopic wavelengths [3]. On the second horn, keeping the world ghost-free is itself a policy with a price schedule: the companion [4] shows that each ghost-removal prescription on record—a boundary condition, a contour, a norm—acts as a survival-conditioning of the underlying process, with a closed-form cost that in the flagship case collapses exactly at the point where the exorcism is most needed. On the third horn the price is the absence of a sample space: there is no stochastic process on definite configurations at all, and hence no single actual history.

The placement. Everett sits on the third horn—and, and this is the point of this section, sits there *comfortably.* Dropping definite configurations is not something the trilemma does *to* Everett; it is what Everett walked in the door already having done. The Everettian never wanted a sample

Horn (what is dropped)	**What is kept**	**The price, as priced in this program**
Empirical Lorentz invariance	coupled ghosts + definite configurations	an observable preferred frame; on the conformal-gravity instance the price forks on the gravitational coupling α_g, both branches costly [3]
Coupled ghosts	empirical Lorentz invariance + definite configurations	every ghost must be exorcised, and every exorcism is a survival-conditioning with a computable cost [4]
Definite configurations	coupled ghosts + empirical Lorentz invariance	no sample space: no closed stochastic process on configurations exists to be had [3]

Table 1: The three horns of the trilemma of [2, 3], with prices. Everett occupies the third row.

space: probability in the Emergent Multiverse enters through decision theory and self-location, not through a stochastic process on configurations [17]. And the reward for paying that price is exactly the good the other horns forfeit: with no beables there is no beable kinematics, no foliation for it to be anchored to, no frame for a ghost sector to expose—manifest covariance survives coupled ghosts unscathed. The trilemma therefore *locates* Everett rather than criticizing it: Everett is the pick-two solution {ghosts, Lorentz}, and its famous virtue and its famous vertigo are revealed as the two faces of a single purchase. The trilemma does not embarrass any horn; it prices each.

An honesty paragraph, to keep the location claim from quietly inflating into an argument. The instantaneity theorem of [2] is *beables-relative*: it says a configuration-space process for a coupled ghost vacuum degenerates unless individuating structure is supplied. Everett has no configuration-space process, so the theorem does not bite Everett at all—which is just the location claim restated from the other side. It is worth being explicit that the trilemma locates Everett by *negation*: nothing in the trilemma's mechanism, which operates on configuration-space processes, reaches Everett at all, and it is Everett's *own* commitment to having no beables that seats it on the drop-configurations horn. The placement is done by Everett's premise, not by any argument of ours. A further dependency we state once and plainly: claims (ii) and (iii) inherit their content from the program's companion papers [2, 3, 4]—the trilemma, the instantaneity theorem, the exposure template, and the exorcism costs are established there, not here—so claim (ii)'s soundness is exactly the soundness of those companions, while claim (iii) rests on them *and*, over and above, on Conjecture 1 (with the comoving-frame and clean-case conditions of Section 5), this paper's own and admittedly conjectural; this paper assembles and prices their results, it does not reprove them. Nothing in this section is evidence against the Everett interpretation, and we do not intend a cumulative-case reading in which a trilemma horn is a demerit. Horns are addresses, not verdicts.

5 The separation: a ghost sector makes it empirical

This is the paper's central section and its one novel claim. The claim is conditional and one of its two supports is a conjecture; we build it in three steps and flag the status of each.

Step 1 (companion results: theorem at the core, argument at field strength). A beables-first field theory is foliation-anchored by construction: defining a configuration space for

field beables requires a choice of spacelike slicing, exactly as in Bohmian field theory [20]. The companion paper [2] sharpens this ancient observation in two ways that matter here. First, the foliation's *determinacy* is a well-defined (quantifiable) functional of the quantum state: the paper exhibits a measurable property of the state quantifying how sharply the state selects a frame, and shows—by exact algebra at the vacuum, and a measured determinacy curve approaching it—that it *vanishes continuously at the vacuum*. A perfect-fluid state with $\rho + p \neq 0$ determines a frame; the vacuum determines none; the boundary $\rho + p = 0$ is the phantom divide. Second—the shield—in ordinary (ghost-free) sectors in quantum equilibrium the anchoring foliation is empirically hidden: no Born-rule observable depends on it. Within the indivisible framework this hiddenness is proven as a no-signaling theorem for the finite-configuration core, and *conjectured*, with a defined proof target, at full field-theory strength [2].

It is this shield that makes our position an extension of Barandes' rather than a heresy within his framework. Barandes' published claims that the formulation is causally local [7, 8] and has no fundamentally preferred foliation [5] concern the causal and signaling structure of quantum theory as instantiated in ordinary matter—and there we agree, on the strength of the same shield: on every sector so far observed, the anchoring foliation is gauge, exactly as unobservable as his theorems require. What follows concerns a sector nobody has observed. We should flag, though, that this reconciliation is our reading and not Barandes'. Restricting "no fundamentally preferred foliation" to ordinary matter, and treating a ghost sector as the regime where the foliation is promoted to observable, is a substantive interpretive move on which Barandes may well disagree: his stated position is that the formulation has no fundamentally preferred foliation, without a matter-content proviso. We claim compatibility with the letter of his theorems, which concern observed sectors, not his endorsement of the extension.

Step 2 (this paper's step, and it is conjectural). A coupled ghost sector would break the hiding. The mechanism is the first horn's exposure result run in reverse: a coupled ghost's decay products are isotropic in the individuating frame and in no other, so a detector moving with velocity β relative to that frame sees a Compton–Getting kinematic dipole aligned with (on the natural identification of the individuating frame with the comoving frame) the CMB dipole, together with a direction-modulated spectral edge $\Lambda(\theta) \simeq \Lambda(1 + \beta\cos\theta)$ at the sector's cutoff. The companion paper computed the template quantitatively for the phantom instance [2]: count-rate dipole amplitudes of a few $\times 10^{-3}$ at $\beta = 1.23 \times 10^{-3}$, roughly 4.0–5.5 $\times 10^6$ signal photons for a 5σ dipole detection with the apex fixed by the CMB dipole, and an edge modulation at the 10^{-3} level recoverable through realistic instrument resolution. The discriminant against mundane comoving backgrounds, which share the Compton–Getting dipole, is the edge modulation together with the absence of source correlation. One load-bearing step here is an assumption, not a derivation, and we flag it as such: the identification of the individuating frame with the comoving (CMB) frame. The instantaneity result of [2] fixes only that *some* frame individuates the decay channels; that this is the frame in which the cosmic matter distribution is at rest is an assumption we make—plausible for a cosmologically prepared sector—but do not derive. It is worth being explicit about which rescue underlies this quantitative template: it is computed for a *frame-cutoff* instance—a momentum cutoff imposed in the individuating frame, which we identify above with the comoving (CMB) frame—rather than for a purely state-functional foliation. The Compton–Getting dipole is generic to any individuating frame, but the spectral edge, and the saturation bound $\Lambda \lesssim 19\,\mathrm{MeV}$ that makes the photon counts feasible, both presuppose that cutoff. By the analysis of Step 3 below, the template therefore already sits in the *confounded* branch; and because a momentum cutoff is a law-level Lorentz-violating completion, it sits in the branch a

no-slicing interpreter may adopt only at the price of manifest covariance, not in the birth-surface branch available for free. It should not be read as a signature of the clean case. If the individuating frame were some other frame, the dipole apex would move with it and the CMB-locked template would need re-pointing; a frame-locked signal would still exist, but its specific direction would not be predicted. We emphasize that these numbers live *inside* the conditional built in this section: they quantify the template that becomes operative only if a coupled ghost sector exists and the conjecture below holds.

We state the logical status of this step with maximal bluntness, because it is the paper's load-bearing conditional. The observability claim is the *converse* of the shield: the shield says ordinary sectors hide the foliation; the converse says a ghost sector exposes it. In [2] the shield itself is a theorem only for the finite-configuration core, and the ghost-induced observability is argued at tree level, not proven. If the full-strength shield were to fail, the separation below would be confounded (foliation effects might leak elsewhere); if the tree-level exposure argument were to fail under resummation, the signal might be absent even on the foliation-anchored reading. We therefore record the step as a conjecture, whose resolution is the natural next proof target for the program:

Conjecture 1 (shield converse; conditional)**.** *In a beables-first indivisible field theory whose matter content includes a ghost sector coupled to ordinary matter, some Born-rule observable of the ghost sector's decay products depends on the anchoring foliation with nonzero effect (generically—away from the phantom divide $\rho + p = 0$, where the foliation-determinacy functional vanishes)—the kinematic dipole and spectral edge of [2] being the template—while all observables of ghost-free sectors remain foliation-independent.*

Two conjuncts are bundled here for economy, but only the first is load-bearing for the falsification below. The null test uses the *exposure* conjunct alone—beables ⇒ a ghost-sector signal, so a null ⇒ not beables, as the template realizes it (Conjecture 1 delivers a *nonzero* effect; reaching a null's full force needs the template's saturating-cutoff and comoving-frame auxiliaries, Step 2 above); the second conjunct (the *shield* for ghost-free sectors, already a theorem for the finite core) governs only whether foliation effects might leak into ordinary sectors and confound the test, not whether the null falsifies. Keying the falsification to exposure alone keeps it from being hostage to the field-strength shield.

Step 3 (the separation, and where its positive direction is confounded). A no-preferred-slicing ontology predicts no such signal *when the ghost Lagrangian is held fixed and Lorentz-invariant and no individuating completion or birth-surface initial condition is invoked*, because then it has nothing for the signal to reveal. On the Everettian reading of the same Lorentz-invariant field content there is no beable kinematics, hence no anchoring foliation, hence no frame for the decay products to lock to; the natural Everettian completions of such a ghost sector are precisely the definition-level, covariance-preserving quantizations that the trilemma's third horn comprises—and those have no frame-locked decay signal to give.

The positive direction, however, carries a confound we must state plainly, because the trilemma's own exposure mechanism supplies it. In Section 4 the individuating structure that rescues a coupled ghost vacuum is furnished, in every rescue the program could find, by one of two kinds of structure, and it is worth keeping them apart. The first is a physically *Lorentz-violating* completion at the level of the law—a momentum cutoff or ghost condensation. The second is a *birth-surface* (initial-condition) structure such as a born vacuum, which is not a law-level Lorentz violation at all but a cosmological initial datum; though sometimes grouped with the Lorentz-violating

rescues, it is more accurately classed as an initial condition. Both routes are available to a no-slicing (Everettian) interpreter. The Lorentz-violating completion can simply be adopted; and the birth surface is available *for free* to any Everettian who accepts big-bang cosmology, so if a bare cosmological initial condition suffices to individuate the decay channels, the confound is *generic*—essentially every realistic Everettian can then reproduce the same signal at no cost, and the positive direction is weaker still. Either way such an interpreter can adopt the identical structure and predict the *same* frame-locked dipole and edge: in that case the signal tracks the completion or the initial condition, not the ontology. A positive detection therefore does *not* entail a foliation-anchored ontology. The separation is clean only in the restricted case where the individuating frame arises *solely* as a state-functional foliation—the covariant-foliation-from-the-wave-function picture of [20], extracted from the state with the Lagrangian held fixed and Lorentz-invariant—rather than from a fundamentally Lorentz-violating completion or a birth surface. Wherever such a completion or initial condition is in play, a positive signal is confounded between the two classes. This clean case should therefore be named as a *third, separate condition* on the separation, alongside the existence of a coupled ghost sector and Conjecture 1: that a state-functional foliation *suffices* to individuate the decay channels. And we must concede that this condition has never been met on paper. No purely state-functional rescue of a ghost vacuum has ever been exhibited: by this program's own survey (Section 4), every known rescue supplies a frame or a birth surface—a momentum cutoff, ghost condensation, a born vacuum—and the [20] construction is a picture for ordinary Bohmian matter, not a demonstrated rescue of the *ghost* vacuum. Conjecture 1, whose antecedent presupposes only a beables-first ghost field theory, does not cover this gap. On current knowledge, therefore, every concrete realization of the positive signal lies in the confounded branch, and the clean case is conjectural *over and above* Conjecture 1. Moreover the birth-surface route is not a gap that further modeling might simply close: an FRW birth surface is generic to any big-bang cosmology, so *if* a bare cosmological initial condition suffices to individuate the decay channels, the confound on the positive direction is not merely a present lacuna but arguably *structural*—unobservable in principle on a realistic cosmological background, not just so-far-unrealized. This strengthens, rather than weakens, the null as the discriminator: if every *viable* beables completion must carry an individuating structure, then every viable beables completion predicts a signal, so the absence of one *at the template's sensitivity* presses hard against every template-saturating realization of the class. The discriminating power is thus asymmetric, and the asymmetry is what keeps claim (iii) non-vacuous. Given Conjecture 1 and the program's survey, the beables class *entails* a nonzero signal, so for an established sector whose computed template puts that signal at the null's sensitivity a null result *falsifies* that class *as the template realizes it*—a nonzero but sub-threshold effect (a sector the template does not cover, *or* a covered sector realized off the comoving frame or below the template's signal fraction) would survive it. The Everettian, by contrast, merely *permits* the signal: the quantization is a free choice—covariance-preserving with a Lorentz-invariant Lagrangian and no completion, or signal-matching by borrowing a completion or birth surface—so Everett entails nothing and remains compatible with *both* outcomes. A positive detection therefore refutes nothing on the no-slicing side, whereas a null—at the template sensitivity, for an established sector—falsifies the beables class (under Conjecture 1); only the beables direction carries an entailment for an experiment to break.

Even a confounded positive, moreover, is not always without force for the no-slicing side—but here we must scope the point carefully, because the two confounding routes are not on a par. A no-slicing interpreter who adopts a fundamentally *Lorentz-violating* completion at the level of the law in order to match the signal thereby forfeits *manifest covariance*—which Sections 1 and 4

identify as exactly the good that dropping configurations was supposed to buy. In that branch a confounded positive fails to select the ontology, yet it still exacts a real and nontrivial price from the horn whose entire purchase was covariance. The birth-surface branch is different, and we concede it: a Lorentz-invariant law with a non-invariant cosmological initial state (an FRW birth surface) forfeits no manifest covariance at the level of the law, so there the individuating structure is available *for free*—as noted above—and a confounded positive costs the Everettian nothing. This law-level covariance does not, however, put a born-vacuum beables theory in the trilemma's forbidden "all three" corner: the trilemma's second element is *empirical* Lorentz invariance—no observationally accessible preferred foliation (Claim 1)—and a born vacuum (conditionally, at tree level) exposes the frame observationally, so a beables theory with a Lorentz-invariant law but a frame-exposing birth surface still drops the empirical-Lorentz-invariance horn. The law-level covariance it retains is not the currency the trilemma trades in. The one branch so far realized as a concrete signal template is, however, the *frame-cutoff* branch, which *is* a law-level Lorentz-violating completion; so the covariance price is one the exhibited realization actually charges. It is the birth-surface branch—the branch that would exact no such price, and whose availability for free is what makes the confound generic—that has not so far been realized as a concrete template. Putting the steps together, conditionally, and with the confound flagged:

> *If* nature contains a coupled ghost sector, *and if* Conjecture 1 holds, *then*: a frame-locked dipole-plus-edge signal in that sector's decay products is *consistent with*—but does not select—the foliation-anchored single-world class, being *confounded* by any fundamentally Lorentz-violating completion or birth-surface initial condition of the sector, either of which reproduces the same signal on a no-slicing ontology—the birth surface at no cost, the Lorentz-violating completion at the price of manifest covariance; the signal indicates foliation-anchoring cleanly only under the further, so-far-unrealized condition that the individuating frame arises purely as a state-functional foliation with the Lagrangian held Lorentz-invariant. A null result, with the ghost sector independently established, is the cleaner discriminator, precisely because the entailment runs one way: the beables class *entails* a signal (of the computed template amplitude, for an established sector, under the assumed comoving-frame identification—a realization off that frame, below the template's signal fraction, or individuated by a completion other than the template's frame cutoff, survives the pointed null), so its absence is evidence against definite configurations and presses toward the no-slicing class, whereas a no-slicing ontology merely *permits* the signal, so a positive leaves it untouched.

Neither outcome selects an interpretation. The positive outcome does not distinguish Bohmian from stochastic beables (Section 6); the null outcome does not distinguish Everett from any other foliation-free reading. What the experiment discriminates is a *class boundary*—slicing-anchored versus not—which is, we believe, precisely the boundary the interpretation debate has always treated as empirically inaccessible. And the Born rule is intact on both sides throughout: the two classes differ not about any probability rule but about what the world contains for the rule to apply to, and the disagreement becomes observable only because a ghost sector's kinematics is (conditionally) sensitive to the difference.

The ancestry, and the contrast that defines the claim. The idea that a hidden-variable ontology might escape the "Everett in denial" charge by empirical divergence is Valentini's: in quantum non-equilibrium, pilot-wave theory diverges observably from quantum mechanics, so

the corpuscles are not idle wheels [15, 18, 19]. But Valentini's escape requires the Born rule to *fail*—non-equilibrium is by definition a violation of Born statistics, and his signal-locality theorem ties the divergence to that violation. Ours is the ghost-sector generalization of the same escape with the requirement inverted: the discriminator is a sector, not a rule failure. One other equilibrium route to foliation observability has been proposed—Bohmian arrival-time statistics in EPRB configurations [21], a laboratory-scale mechanism resting on the contested measurability of arrival times, and disjoint from the sector-based discriminator here. Beyond it we know of no prior proposal that separates interpretation classes observationally while holding the Born rule fixed. The nearest relatives sharpen the contrast. Dürr, Goldstein, Norsen, Struyve, and Zanghì showed that a Bohmian theory can extract a covariant foliation from the wave function itself, adding that such a foliation is undetectable in quantum equilibrium [20]—our scenario is precisely the loophole in that equilibrium hiding: it is a fact about ordinary sectors, and a ghost sector (conditionally) breaks it while equilibrium holds. Sahin's recent program of quantum-field-theoretic phenomenology for ψ-epistemic models [23] needs deviations from quantum-mechanical predictions to generate signals; we need none. And the fragility results of [22] are companions in spirit: empirical equivalence between interpretations is a regime-bound theorem, not a decree, and our claim is that the regime has a physically meaningful boundary.

The conditional, stated plainly. The trigger may never fire. If nature contains no coupled ghost sector, the separation never becomes operative and the equivalence between the classes stands unbroken—this paper's claim is a conditional falsifiability statement, the same logical DNA as the rest of the corpus. What would count: on the input side, an established phantom sector (a $w < -1$ equation of state confirmed jointly with structure-growth data discriminating a genuine field from a gravitational mimic) is the scenario in which the template of [2] applies as computed; current cosmological data contain disputed hints and nothing more, and we take no position on them here, deferring to the companion's discussion. On the output side, the null is well-defined: ghost sector established, exposure and systematics sufficient for the 10^{-3} dipole and edge, and no frame-locked signal. We emphasize once more that the detection numbers quoted above sit on a knife-edge of signal fraction detailed in [2]; the separation is an experiment nature may offer, not one we can schedule. Absent the sector, the statement is vacuously safe and empirically idle—and we say so.

6 What does not follow

Four disclaimers, each of which we regard as part of the paper's content rather than throat-clearing.

(i) The instantaneity theorem is not an argument against Everett. It is beables-relative (Section 4): it constrains configuration-space processes, and Everett has none. We have used it only to price the horn Everett declines, never to argue against the horn Everett occupies; any reading of this corpus as a cumulative case against the Everett interpretation is a misreading.

(ii) Division events are not branch events—so far as anyone has shown. There is a tempting translation: division events, the isolated times at which an indivisible process factorizes, resemble the decoherence times at which Everettian branch structure stabilizes; and the terminal-indivisibility class of [1]—no division events beyond a finite time—would translate to "no stable branch structure available in ghost regimes." We flag this correspondence as *suggestive translation only.* The program's theorems concern configuration-level divisors; Everettian branching is defined through decoherence functionals on coarse-grained histories; no theorem connecting the two exists, in this corpus or elsewhere to our knowledge. It is a conjecture-shaped hole, not a claim.

(iii) Nothing here selects among single-world readings. The separation of Section 5 discriminates slicing-anchored from non-slicing ontologies. A positive signal would not distinguish Bohmian beables from stochastic beables, nor either from any other foliation-anchored ontology; a null would not distinguish Everett from foliation-free alternatives. The experiment, if nature grants it, adjudicates a class boundary and nothing finer.

(iv) The separation claim is conjectural several times over. It conditions on the existence of a coupled ghost sector (an empirical question, open), and on Conjecture 1 (a theoretical question, open: the shield is proven only for the finite core, and the exposure argument is tree-level); and its *clean* (class-selecting) direction conditions further on a state-functional foliation sufficing to individuate the decay channels—a condition that, on current knowledge, no exhibited rescue meets, so the confounded branch is the only one so far realized. We have tried to write every sentence of Section 5 so that it survives the worst case on all three conditions; where a sentence seems to outrun them, the conditions govern.

7 Discussion

Interpretations of quantum mechanics that differ in a register physics can reach are rare; that has always been the debate's melancholy. The argument of this paper is that the register exists and has a name: ghosts. A coupled ghost sector is the one place the program has found in field theory where, at tree level, the existence of a sample space stops being free—where definite configurations demand an individuating structure and (conditionally) advertise it, and where declining to have configurations converts the signal from an entailment into a mere permission. If the reconstructions of Section 2 restore the underdetermination that the no-alternative argument had claimed to close, the trilemma and the separation say where the underdetermination could end: not in the seminar room, but in the direction-dependence of a diffuse background.

We do not expect the question settled soon, and the paper's architecture is built for that: claim (i) stands as a piece of underdetermination independent of any ghost physics—with two qualifications this paper has been at pains to make, that it is conditional on Barandes' (cited, not here established) exit from the PBR theorem, and that its *defense* against the strongest Everettian reply, the bookkeeping objection, is not fully ghost-independent, since the decisive answer to that reply is the ghost-dependent claim (iii); unconditionally, claim (i) meets the objection only with the defensive parity and disanalogy replies. Claim (ii) stands at tree level whatever the observations do; claim (iii) waits on a sector, a conjecture, and—for its positive direction—an unexhibited clean case. What the trilemma does unconditionally is raise the stakes of the ghost question itself. Whether nature contains coupled ghosts has been, to date, a question about instability and regularization; on the argument here it is also, conditionally, the question of whether the interpretation debate has an empirical future.

We end with the two vertigos, because the corpus should take no side the data cannot take. The Everettian asks us to accept worlds beyond counting, splitting without cease, and offers in exchange a formalism taken perfectly literally. The stochastic reading asks us to accept memory without storage—a world whose single history obeys a law that remembers what no beable records—and offers in exchange a single world. Both offers are coherent; both prices are real; and the argument of this paper is that, if nature is unkind enough to contain a ghost sector, she may also be kind enough to tell us which price she has already paid.

Author contributions and use of AI. This manuscript was researched and written by the large language model Claude Fable 5 (Anthropic), working under the direction of Jonathan

Torkelson, who conceived the program's originating question, directed the sessions, and reviewed the outputs. The author is a professional engineer, not a physicist, and makes no claim to independent verification of the technical content; the manuscript is circulated precisely to obtain such verification from professional physicists. All errors are owned by the process that produced them.

Code availability. This is a synthesis paper and proves no new theorem and runs no new numerics. Every numerical claim quoted here—the divisor computations and terminal-indivisibility constants, the dipole and spectral-edge template and its photon counts, the foliation-determinacy functional—is inherited from the companion papers and reproducible from their public scripts; entry points and the program-wide index are collected at the hub repository https://github.com/embeddetech/ghosts-program.

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